THE REMOVAL OF TRACE POLLUTANTS IN WATER USING ADVANCED OXIDATION PROCESS UVB/BREAKPOINT CHLORINATION

Master Thesis

Institute of Environmental Engineering

National Yang Ming Chiao Tung University

Student: Shao Fang Lin

Advisor: Yi Hsueh Chuang

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Shao Fang Lin

Institute of Environmental Engineering

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ABSTRACT

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Climate change has led to water shortages becoming a global issue. Many studies are focused on finding stable, reliable, and cost-effective alternative water sources to achieve sustainable water supply. Wastewater recycling is one of the solutions to water shortages. By recycling secondary and tertiary effluent from wastewater treatment plants and processing it through a comprehensive Full Advanced Treatment (FAT) system, it can be used as an alternative water source for drinking water facilities.

To address the issues of low photon utilization efficiency and residuals in UV/H2O2 advanced oxidation processes, scholars are now using UV/HOCl as a replacement. However, HOCl undergoes breakpoint chlorination reactions with chloramines present in the RO permeate. This study uses Kintecus software to simulate the concentration changes of free chlorine and chloramines in RO permeate based on the mathematical model developed by Jafvert and Valentine. The average concentration of species after 30 seconds of breakpoint chlorination is multiplied by their molar absorptivity coefficients in the 230-400 nm range to obtain absorbance. Simulation results show that dichloramine has a high absorbance in the UVB range. This suggests that switching the UV/HOCl light source to UVB wavelengths could enhance the efficiency of the UV advanced oxidation process in degrading trace pollutants.

In this study, laboratory-synthesized RO permeate was spiked with six model micropollutants: **nitrobenzene, benzoic acid, carbamazepine, caffeine, 1,4-dioxane, and N,N-diethyl-m-toluamide**. The objective was to investigate whether UVB306 nm/breakpoint chlorination can effectively transfer energy to dichloramine, thereby generating free radicals more efficiently and enhancing pollutant degradation. Finally, the study further explored optimizing the operating conditions of UVB306 nm/breakpoint chlorination advanced oxidation processes (AOP) by examining the effects of a 90-second dark reaction post-chlorination before irradiation, reducing the energy output per unit time of the UVB306 nm lamp, and changing the reactor's reflective surface material. Additionally, the study examined the formation and degradation of **N-nitroso dimethylamine (NDMA).**

The study found that the degradation efficiency of pollutants was optimal using a 4 W UVB306 nm/breakpoint chlorination system in a Teflon reactor. Compared to using an 8 W UVC254 nm/H2O2 and an 8 W UVC254 nm/breakpoint chlorination in a stainless steel reactor, the degradation efficiencies for nitrobenzene and 1,4-dioxane were improved by

73% and 29%, respectively. In the degradation of NDMA, the 4 W UVB306 nm/breakpoint chlorination with a Teflon reactor showed promising results. For countries facing water scarcity, the UVB with breakpoint chlorination technique in advanced oxidation processes holds potential. Future studies need to ensure that the treated effluent does not exhibit higher toxicity.

Keywords: UVB306 nm/breakpoint chlorination, synthetic RO permeate, model micropollutant, absorbance, N-nitroso dimethylamine

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CHAPTER 1. RESEARCH BACKGROUND

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1.1 Research Background and Motivation

Environmental change and climate change, population growth, and economic development have caused water scarcity to become a global problem. Taiwan, California in the United States, Australia, and Singapore all face water shortages. In California, for example, climate change has caused the snowpack to gradually decrease (Ullrich et al., 2018). Therefore, finding stable, reliable, and cost-effective alternative water sources is currently a major direction and goal of research.

Wastewater recycling is part of the concept of sustainable water supply and "closing the water loop," and is one means of addressing water scarcity. In California, secondary and tertiary effluent from municipal wastewater treatment plants is collected and treated through a full advanced treatment (FAT) system (Soufan et al., 2013). After stabilization, the treated water can be supplied to drinking-water plants as an alternative source (Gerrity et al., 2013; Kwon et al., 2020).

As shown in Figure 1-1, the **FAT system first adds hypochlorous acid (HOCl) before membrane treatment processes** such as microfiltration (MF), ultrafiltration (UF), and reverse osmosis (RO) to inhibit biological growth on the membranes (Wang et al., 2021). Because secondary and tertiary effluents contain some ammonia nitrogen, added HOCl reacts with ammonia nitrogen and forms chloramine species, mainly monochloramine and dichloramine**. Because chloramine species are small neutral molecules that are not easily removed by membranes (Gerrity et al., 2013), they ultimately remain in the RO permeate and enter the UV reactor**. The conventional approach is to use H2O2 as the oxidant for advanced oxidation treatment in the UV reactor.

Figure 1-1. FAT treatment flow for wastewater reuse. Source: Gerrity et al. (2013).

Although UV/H2O2 is widely used in water-reuse treatment systems internationally, it has two major problems. First, at 254 nm, chloramine species have much higher molar absorptivity than the 18.6 M^-1 cm^-1 of H2O2 (Yin et al., 2020). **If the background matrix contains a certain concentration of chloramines, most of the UV energy emitted into the water is absorbed by chloramines, leaving very little energy for H2O2. Studies indicate that only about 10% of H2O2 consumed in a UV/H2O2 AOP is used to remove pollutants in water** (Gao et al., 2020). Second, before treated water enters an environmental buffer water body, post-chlorination is required to maintain disinfection. To maintain the residual chlorine concentration, additional HOCl must be added to react with and remove the remaining H2O2 (Equation 1.1) (Guo et al., 2017; Kwon et al., 2020; Roback et al., 2021; Wu et al., 2016).

HOCl + H2O2 -> O2 + H2O + Cl- + H+ (1.1)

To address the low photon-utilization efficiency and residual-oxidant problems of UV/H2O2, many researchers over the past decade have studied UV/HOCl or UV/chloramine as alternatives. Kwon et al. (2020) and Roback et al. (2021), for example, investigated RO permeate from the Orange County Water District Groundwater Replenishment System (OCWD/GWRS) and examined practical applications of UV/HOCl and UV/chloramine in place of UV/H2O2.

Because chloramines are inevitably present at some concentration in RO permeate from water-reuse systems, when the advanced oxidation step after the RO membrane is changed to UV/HOCl, HOCl reacts with chloramines and, at certain chlorine-to-nitrogen molar ratios, produces a breakpoint chlorination reaction. The final forms produced by the reaction between HOCl and chloramines depend mainly on the molar ratio of free chlorine (HOCl and OCl-) to ammonia and the reaction time (Aghdam et al., 2021).

In general, under neutral conditions, breakpoint chlorination reaches a stable state approximately 20-30 minutes after the reaction begins and can be divided into three stages

(Figure 1-2). Before a chlorine-to-nitrogen mass ratio of 5:1 (approximately Cl2/N molar = 1:1), when free chlorine is added to water containing ammonia nitrogen, free chlorine undergoes a series of complex reactions with ammonia to form combined residual chlorine, primarily monochloramine. As the amount of free chlorine increases, combined residual chlorine and total chlorine concentrations also increase. When Cl2/N molar reaches 1:1, free chlorine begins consuming the combined residual chlorine until the breakpoint is reached, at which point Cl2/N molar is approximately 1.5:1-2:1 (Metcalf et al., 2007; Wahman, 2018).

The breakpoint is the point at which the total chlorine concentration in the water is lowest. Before the breakpoint, dichloramine is the major species. When the chlorine-to-nitrogen molar ratio exceeds the breakpoint, the combined residual chlorine has been consumed and further addition of free chlorine remains in the water as free residual chlorine (Jafvert and Valentine, 1992).

Figure 1-2. Breakpoint chlorination curve. Source: Metcalf et al. (2007).

At the microscopic reaction level, based on the reaction mechanism of free chlorine and ammonia nitrogen described by Jafvert and Valentine (1992), NH3 reacts with HOCl to form NH2Cl (Equation 1.2). NH2Cl can then react with HOCl to form NHCl2 (Equation 1.3), or undergo a disproportionation reaction to form NHCl2 (Equation 1.5). NHCl2 can subsequently react with HOCl to form NCl3 (Equation 1.4), or hydrolyze with water to form an intermediate product. NH2Cl or NHCl2 can then react with the intermediate to regenerate HOCl and other byproducts (Equations 1.6 and 1.7). Byproducts include N2, H2O, Cl-, H+, and NO3- (Equation 1.8). NCl3 can also react with NH2Cl or NHCl2 to regenerate HOCl (Equations 1.9 and 2.0), while HOCl further oxidizes NHCl2 to NCl3, sustaining the catalytic reaction.

NH3 + HOCl -> NH2Cl + H2O (1.2)

NH2Cl + HOCl -> NHCl2 + H2O (1.3)

NHCl2 + HOCl -> NCl3 + H2O (1.4)

NH2Cl + NH2Cl + H+ -> NHCl2 + NH3 + H+ (1.5)

NH_2Cl + intermediate I -> products (1.6)

$NHCl_2$ + intermediate I -> HOCl + products (1.7)

$2HOCl + NHCl_2 + H_2O \rightarrow NO_3^- + 5H^+ + 4Cl^-$ (1.8)

$NCl_3 + NH_2Cl \rightarrow$ HOCl + products (1.9)

$NCl_3 + NHCl_2 \rightarrow HOCl + HOCl$ + products (2.0)

Aghdam et al. (2021) noted that HOCl has a higher quantum yield and molar absorptivity and can therefore generate radicals more effectively during UV/HOCl AOP. The same paper noted that ammonia nitrogen is commonly present in rivers, wastewater, and reclaimed water. Municipal wastewater and reclaimed wastewater may contain approximately 1-25 mg/L ammonia nitrogen, and adding different amounts of HOCl to water containing ammonia produces chloramine species (NH_2Cl, $NHCl_2$, and NCl_3) (Aghdam et al., 2021).

When UV/HOCl AOP is used to treat water containing ammonia nitrogen, NH_2Cl forms rapidly when Cl_2/N molar < 1, making the process effectively UV/NH_2Cl AOP. When excess HOCl is added (Cl_2/N molar > 1.5-2), breakpoint chlorination occurs (Aghdam et al., 2021). However, many studies of UV/breakpoint chlorination have not considered the dynamic changes in chlorine and chloramine speciation during the process or whether irradiation times match the reaction time for breakpoint chlorination. Therefore, differences between UV/HOCl and UV/breakpoint chlorination in practical applications cannot always be explained.

Zhang et al. (2019b) found that RO permeate from a California water-reuse plant contained approximately 4 mg/L as Cl_2 of chloramines, including approximately 2.5 mg/L as Cl_2 (35.21 uM) NH_2Cl and 1.5 mg/L as Cl_2 (10.56 uM) $NHCl_2$. A normal UV/HOCl operating dose at a full-scale facility is 5-8 mg/L as Cl_2 (70.4-112.7 uM) HOCl (Kwon et al., 2020; Zhang et al., 2019b). Kintecus simulations using the Jafvert and Valentine model showed that when 84.5 uM (~6 mg/L) HOCl was added to RO permeate under dark conditions, HOCl and NH_2Cl rapidly decreased over approximately 30 seconds to 1 minute while $NHCl_2$ was generated in large quantities. Around 3 minutes after the reaction began, the species concentrations approached a stable state (Figure 1-3). Thus, $NHCl_2$ was the main product after breakpoint chlorination in the simulated RO permeate.

Figure 1-3. Simulated changes in species concentrations over five minutes after adding 84.5 uM HOCl to pH 5.8 RO permeate containing 35.21 uM NH2Cl and 10.56 uM NHCl2.

Full-wavelength analysis of NH2Cl, NHCl2, NCl3, HOCl, and OCl- showed that each species has a different absorbance profile. Because the first five minutes of breakpoint chlorination are the period when chloramine concentrations change most rapidly, clarifying the radical-generation rates of the different oxidant species during this period is important for improving UV AOP efficiency. The average concentration of each species at 30 seconds was multiplied by its molar absorptivity from 230-400 nm to obtain absorbance (Figure 1-4). At 254 nm, all species generally showed low absorbance. Around 290-300 nm, however, NHCl2 showed high absorbance of about 0.011 cm^-1, indicating that NHCl2 was the dominant absorbing species in this range.

Figure 1-4. Kintecus simulation of absorbance based on the Jafvert and Valentine model at 30 seconds of breakpoint chlorination, using the average concentrations of HOCl, OCl-, NH2Cl, NHCl2, and NCl3 multiplied by their molar absorptivities over 230-400 nm. Simulation conditions: pH 5.8; [NH2Cl] = 35.21 uM; [NHCl2] = 10.56 uM.

However, UV/HOCl can increase the formation of nitrosamines. Schreiber and Mitch (2007) used dimethylamine (DMA) as a model precursor and observed N-nitrosodimethylamine (NDMA) formation at different Cl2/N molar ratios during breakpoint chlorination. NDMA concentration increased rapidly between 1.5 < Cl2/N molar < 2 and was about six times higher than at Cl2/N molar < 1.5 (Figure 1-5). Previous research by Bolton and Stefan found that RO treatment removes only about 30% of NDMA (Szczuka et al., 2020). However, NDMA has a high molar absorptivity at 254 nm, reaching 1650 M^-1 cm^-1. Thus, although UV254 nm/breakpoint chlorination can promote NDMA formation, UV254 nm can also photolyze NDMA effectively (Soltermann et al., 2013).

Figure 1-5. Relationship between Cl2/N molar ratio during chlorination disinfection and NDMA concentration. Source: Schreiber and Mitch (2007).

1.2 Research Objectives

According to Aghdam et al. (2021), in a conventional semi-enclosed laboratory reactor, breakpoint chlorination in the dark produced little additional pollutant degradation after about five minutes (Figure 1-6a). Researchers then investigated UV/breakpoint chlorination for benzoic acid. Figure 1-6b shows that 35-40 minutes of irradiation increased benzoic-acid removal from 20% to 70%. One possible reason is the geometry of the conventional laboratory-scale semi-enclosed reactor. To produce partially collimated light, a distance is maintained between the UV source and the water surface (Figure 1-7), reducing the light intensity reaching the water and requiring a longer irradiation period. More importantly, the results may depend on how long different species have been reacting, which can change oxidant forms and the composition of disinfection byproducts (DBPs).

Kwon et al. (2020) reported that a pilot-scale AOP system can provide approximately 700-1000 mJ/cm2 of UV dose within about one minute, which more closely matches breakpoint chlorination reaction times and practical hydraulic retention times in treatment plants.

Figure 1-6. Effect of initial chlorine dosage on benzoic-acid degradation during (a) breakpoint chlorination in the dark and (b) UV254 nm/breakpoint chlorination. Conditions: pH 7; [benzoic acid]0 = 5 uM; [N-NH3]0 = 1 mg/L; [HOCl]0 = 7.6, 9.6, 12.6, and 17.6 mg/L as Cl2. Source: Aghdam et al. (2021).

Figure 1-7. Schematic of a conventional semi-enclosed laboratory-scale reactor.

Based on these findings and the simulations, the study proposed two hypotheses. Hypothesis 1: changing the light source from the commonly used 254 nm wavelength to a UVB source around 300 nm could transfer energy more effectively to dichloramine, allowing more efficient radical generation and improved pollutant degradation. Hypothesis 2: compared with UV/HOCl and UV/chlorine, using a UVB light source together with breakpoint chlorination as an AOP in a water-reuse treatment system may reduce NDMA and other nitrogenous disinfection byproducts.

This study therefore investigates pollutant degradation using UVB306 nm with breakpoint chlorination, examines whether UVB irradiation promotes radical generation by dichloramine, and compares the process with UV254 nm/breakpoint chlorination and

UV/H2O2. The study also seeks optimized operating conditions for the UVB306 nm/breakpoint chlorination AOP and examines NDMA formation and degradation under the optimized conditions.

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CHAPTER 2. LITERATURE REVIEW

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2.1 Wastewater Reuse

2.1.1 Purpose and Methods of Wastewater Reuse

Recycling and reusing wastewater can reduce demand for freshwater resources, relieve water stress, and provide sustainable water supplies in water-scarce regions. It can also reduce withdrawals from rivers, lakes, and groundwater, thereby helping protect aquatic ecosystems and improve water quality.

The main objectives of wastewater reuse can be achieved with a full advanced treatment (FAT) process (Gerrity et al., 2013). Wastewater is treated using membranes and advanced oxidation as physical and chemical barriers and is then, depending on the reuse purpose, discharged to engineered buffer reservoirs or environmental buffer water bodies, thereby improving contaminant removal and water quality.

2.1.2 Wastewater Treatment Process in a FAT System

As shown in Figure 2-1, the main units of the FAT system are membrane processes such as MF/UF and RO, together with a UV/H2O2 advanced oxidation process. MF/UF and RO use physical filtration to remove particles, microorganisms, and dissolved substances. The principal technologies are described below.

1. Microfiltration (MF):

Microfiltration is a physical filtration process that directly filters particles, bacteria, suspended solids, and large organic molecules through micropores. Typical pore sizes are 0.1-10 micrometers, so MF cannot effectively remove smaller dissolved substances.

2. Ultrafiltration (UF):

Ultrafiltration is a further physical filtration process capable of effectively removing microorganisms, viruses, organic molecules, colloids, and macromolecular substances while retaining dissolved inorganic substances. Typical UF pore sizes are 0.001-0.1 micrometers, smaller than those of MF. UF is widely used in drinking-water treatment, industrial water treatment, and wastewater reuse to provide high-quality water.

3. Reverse Osmosis (RO):

Reverse osmosis is a pressure-driven membrane filtration process. Water is forced under high pressure through a semipermeable membrane, removing dissolved solutes, salts, organic compounds, and microorganisms. RO is not effective at removing small neutral molecules (Gerrity et al., 2013). RO membrane pores are typically about 0.0001-0.001 micrometers, allowing removal of almost all dissolved substances and microorganisms. RO is widely used for drinking-water treatment, seawater desalination, industrial water, and wastewater reuse to produce high-quality water with low dissolved-solids content.

In a typical FAT wastewater-reuse system, secondary and tertiary effluent is collected and treated with a high concentration of HOCl. Ammonia in the effluent reacts with added HOCl to form chloramines, mainly monochloramine and dichloramine, thereby controlling biological growth in downstream membrane processes. Because small neutral molecules are not easily removed by RO, chloramines ultimately appear in the RO permeate and enter the UV reactor. Conventionally, H_2O_2 is used as the oxidant in the UV reactor for advanced oxidation. After stabilization, the treated reclaimed water can be supplied to a water treatment plant as an alternative source.

Figure 2-1. FAT treatment flow for wastewater reuse. Source: Gerrity et al. (2013).

2.2 Advanced Oxidation Processes

Advanced oxidation processes (AOPs) are advanced technologies used in water and wastewater treatment, primarily to remove organic pollutants, heavy metals, and other harmful substances. AOPs generally use oxidants to alter or destroy pollutant molecular structures, converting them into harmless substances or forms that are easier to treat (Fang et al., 2014). Oxidants can generate radicals more efficiently under irradiation, creating processes such as UV/H2O2 and improving contaminant removal. AOPs are widely used in industrial wastewater treatment, drinking-water treatment, and groundwater remediation.

2.2.1 Characteristics of Advanced Oxidation Processes

Three common characteristics of AOP technologies are: (1) AOPs use highly reactive oxidants such as ozone (O3) and hydrogen peroxide (H2O2) to oxidize or destroy the chemical structures of organic pollutants; (2) AOPs may use ultraviolet or visible light to activate oxidants and promote pollutant degradation; and (3) ozonation itself is a form of AOP in which ozone gas oxidizes contaminants in water and converts them into compounds that are easier to treat.

Based on previous UV/H2O2 studies, the effectiveness of an AOP depends in part on the steady-state concentration of radicals in the system. After radicals are generated, they react with substances in the system and are continually generated and consumed, creating a dynamic balance that maintains a stable concentration (Chuang and Shi, 2022). Equation 2.1 shows that radical steady-state concentration depends on the oxidant molar absorptivity (epsilon), quantum yield (Phi) (Yin et al., 2020), pollutant concentration, and intermediates. If the radical production rate exceeds the radical scavenging rate, radical steady-state concentration increases and pollutant removal can improve.

[R•]ss = radical production rate / radical scavenging rate

= [E0 * FS(lambda) * FOxi(lambda) * Phi(lambda)] /

[koxidant[oxidant] + kc[contaminant] + sum(ki[Si])]

(2.1)

Here, [R•]ss is radical steady-state concentration; E0 is the surface light intensity; FS(lambda) is the fraction of incident light absorbed by the solution; FOxi(lambda) is the fraction of light absorbed by the oxidant; Phi(lambda) is the quantum yield; koxidant is the reaction rate constant between the oxidant and radical; and kc is the reaction rate constant between the pollutant and radical.

2.2.2 Problems with UV/H2O2

UV/H2O2 is one of the most commonly used AOP methods, but over the past decade many problems have been identified. First, before membrane treatment in a FAT system, secondary and tertiary effluents receive a small amount of HOCl as a biocide. HOCl reacts with ammonia in the background matrix to form chloramines. Further analysis shows that at 254 nm, chloramines have a molar absorptivity of about 352 M^-1 cm^-1, much greater than the 18.6 M^-1 cm^-1 of H2O2 (Yin et al., 2020) (Figure 2-2). Thus, when both H2O2 and chloramines are present in the UV reactor, most UV energy is absorbed by the chloramines and little is available to H2O2. Studies indicate that only about 10% of H2O2 consumed in a UV/H2O2 AOP is used to remove pollutants (Gao et al., 2020).

Second, treated effluent requires post-chlorination before entering an environmental buffer water body to maintain disinfection. Additional HOCl is needed to maintain residual chlorine and react with remaining H2O2 (Equation 2.2) (Guo et al., 2017; Kwon et al., 2020; Roback et al., 2021; Wu et al., 2016), which increases overall operating cost.

HOCl + H2O2 -> O2 + H2O + Cl- + H+ (2.2)

Figure 2-2. Comparison of molar absorptivity of H2O2 and chloramines.

Because of the two main problems of low photon-utilization efficiency and oxidant residuals in UV/H2O2, many researchers have studied UV/HOCl and UV/chloramines as alternatives (Kwon et al., 2020; Roback et al., 2021). A major reason is that photolysis of HOCl can generate highly reactive radicals such as •Cl and •OH (Equations 2.3-2.5).

HOCl + hnu -> •OH + •Cl (2.3)

OCl- + hnu -> •O- + •Cl (2.4)

•O- + H+ -> •OH (2.5)

2.3 Breakpoint Chlorination Reaction

2.3.1 Reaction Mechanism of Breakpoint Chlorination

Free residual chlorine refers to active chlorine species existing in water as hypochlorous acid (HOCl) and hypochlorite ion (OCl-) (Equations 2.6-2.8). Their relative proportions depend on pH; at lower pH, the proportion of HOCl is higher. Both HOCl and OCl- are strong oxidants and can damage microbial cell membranes and intracellular proteins and other biological macromolecules.

Cl- + H2O -> HOCl + OH- (2.6)

Cl2 + H2O -> HCl + HOCl (2.7)

HOCl -> H+ + ClO- (2.8)

When free ammonia (ammonium [NH4+] and ammonia [NH3]) and free residual chlorine (HOCl and OCl-) coexist in water, three inorganic chloramines can form: monochloramine (NH2Cl), dichloramine (NHCl2), and trichloramine (NCl3) (Equations 2.9-2.11).

NH3 + HOCl -> NH2Cl + H2O (2.9)

NH2Cl + HOCl -> NHCl2 + H2O (2.10)

$NHCl_2 + HOCl \rightarrow NCl_3 + H_2O$ (2.11)

Secondary and tertiary effluents in water-reuse processes contain free ammonia. If water contains a certain amount of ammonia nitrogen and different amounts of free residual chlorine are added according to the chlorine-to-nitrogen molar ratio (Cl_2/N molar), the breakpoint chlorination curve shown in Figure 2-3 can be obtained after approximately 30 minutes.

Breakpoint chlorination generally consists of three stages. Before a chlorine-to-nitrogen mass ratio of 5:1 (Cl_2/N molar = 1:1), free residual chlorine reacts with ammonia nitrogen through a series of complex reactions, mainly forming monochloramine. As free chlorine increases, monochloramine and total chlorine concentrations also increase. When Cl_2/N molar reaches 1:1, free residual chlorine begins to consume combined residual chlorine until the breakpoint, at which point Cl_2/N molar is about 1.5:1-2:1 (Metcalf et al., 2007; Wahman, 2018).

The breakpoint is the point at which total chlorine concentration is lowest. Dichloramine is the main species before the breakpoint. After the breakpoint, total chlorine concentration remains relatively stable because ammonia has been consumed and additional free chlorine remains in the water as free residual chlorine (Jafvert and Valentine, 1992).

Figure 2-3. Breakpoint chlorination curve. Source: Metcalf et al. (2007).

The reaction mechanism summarized in Table 2-1 is based on Jafvert and Valentine (1992). NH_3 reacts with HOCl to form NH_2Cl (F1). NH_2Cl can react with HOCl to form $NHCl_2$ (F3) or undergo disproportionation to form $NHCl_2$ (F5). $NHCl_2$ reacts with HOCl to form NCl_3 (F4), or hydrolyzes with water to form an intermediate. NH_2Cl or $NHCl_2$ can react with the intermediate to form HOCl and byproducts (F8, F9); byproducts include N_2, H_2O, Cl^-, H^+, and NO_3^- (F14). NCl_3 can react with $NHCl_2$ or NH_2Cl to form HOCl (F12, F13), and HOCl further oxidizes $NHCl_2$ to NCl_3, allowing the catalytic reaction to continue.

Table 2-1. Reaction pathways of free residual chlorine and ammonia nitrogen in water (Jafvert and Valentine, 1992).

F1	HOCl + NH3 -> NH2Cl + H2O	kU1[HOCl][NH3]	4.17x10^6 M^-1 s^-1
F2	NH2Cl + H2O -> HOCl + NH3	kU2[NH2Cl]	2.10x10^-5 s^-1
F3	NH2Cl + HOCl -> NHCl2 + H2O	kU3[HOCl][NH2Cl]	2.78x10^2 M^-1 s^-1
F4	NHCl2 + H2O -> NH2Cl + HOCl	kU4[NHCl2]	6.39x10^-7 s^-1
F5	NH2Cl + NH2Cl + H+ -> NHCl2 + NH3 + H+	kU5[NHCl2]^2	6.94x10^3 M^-2 s^-1
F6	NHCl2 + NH3 + H+ -> NH2Cl + NH2Cl + H+	kU6[NH3][NHCl2][H+]	6.00x10^4 M^-2 s^-1
F7	NHCl2 + OH- -> I (HNO) + products	kU7[NHCl2][OH-]	1.10x10^2 M^-1 s^-1
F8	NHCl2 + I (HNO) -> HOCl + products	kU8[NHCl2][I]	2.78x10^4 M^-1 s^-1
F9	NH2Cl + I (HNO) -> products	kU9[NH2Cl][I]	8.30x10^3 M^-1 s^-1
F10	NH2Cl + NHCl2 -> products	kU10[NH2Cl][NHCl2]	1.53x10^-2 M^-1 s^-1
F11	NHCl2 + HOCl -> NCl3 + H2O	kU11[NHCl2][HOCl]	5.60x10^-7 M^-1 s^-1
F12	NCl3 + NHCl2 -> HOCl + HOCl + products	kU12[NCl3][NHCl2][OH-]	5.56x10^3 M^-1 s^-1
F13	NCl3 + NH2Cl -> HOCl + products	kU12[NCl3][NH2Cl][OH-]	1.39x10^2 M^-1 s^-1
F14	2HOCl + NHCl2 + H2O -> NO3- + 5H+ + 4Cl-	kU5[HOCl][NHCl2]	2.30x10^2 M^-1 s^-1

2.3.2 Comparison of Radical Generation by Monochloramine and Dichloramine

When free residual chlorine and ammonia nitrogen coexist, breakpoint chlorination occurs. The main sources of radicals are monochloramine and dichloramine. Radical generation depends on the oxidant's molar absorptivity (epsilon) and quantum yield (Phi), and the resulting radical concentration affects pollutant degradation. Molar absorptivity describes the ability of an oxidant to absorb light at a particular wavelength, while quantum yield

describes the number of radicals generated after the oxidant absorbs light (Chuang and Shi, 2022).

According to Yin et al. (2020), the molar absorptivities of NH2Cl at 255, 265, 285, and 300 nm are 345, 174, 75, and 4 M^-1 cm^-1, respectively. The corresponding values for NHCl2 are 96, 111, 214, and 261 M^-1 cm^-1. Table 2-2 gives apparent and intrinsic quantum yields. Because radical generation depends on both molar absorptivity and quantum yield, their product provides a better comparison of radical generation at a given wavelength. The results in Table 2-3 show that at 255 nm, NH2Cl has a higher product of intrinsic quantum yield and molar absorptivity than NHCl2. At 265, 285, and 300 nm, NHCl2 has the higher product. Thus, 255 nm favors radical generation from NH2Cl, while the 280-320 nm UVB range favors radical generation from NHCl2. Therefore, the primary objective of this research is to create more dichloramine in the system so that dichloramine can generate large quantities of radicals under UVB irradiation, thereby improving UV AOP performance and ultimately enabling application in water-reuse drinking-water systems.

Figure 2-4. Absorption spectra and molar absorptivities of NH2Cl and NHCl2 at the tested wavelengths. Source: Yin et al. (2020).

Table 2-2. Apparent and intrinsic quantum yields of NH2Cl and NHCl2 (Yin et al., 2020).

Wavelength (Mangalgiri et al.) | Phi_app-NH2Cl | Phi_app-NHCl2 | Phi_inn-NH2Cl | Phi_inn-NHCl2

255 (257.7) | 0.54 +/- 0.02 | 1.99 +/- 0.05 | 0.34 +/- 0.01 | 0.93 +/- 0.02

265 (268) | 0.50 +/- 0.01 | 1.84 +/- 0.03 | 0.32 +/- 0.01 | 0.84 +/- 0.01

285 (282.3) | 0.47 +/- 0.02 | 1.44 +/- 0.01 | 0.31 +/- 0.01 | 0.65 +/- 0.03

300 (301.2) | 0.39 +/- 0.04 | 1.22 +/- 0.04 | 0.27 +/- 0.02 | 0.52 +/- 0.04

Table 2-3. Product of intrinsic quantum yield and molar absorptivity (Yin et al., 2020).

255 nm: NH2Cl 117.3 > NHCl2 89.28

265 nm: NH_2Cl 55.68 < $NHCl_2$ 93.24

285 nm: NH_2Cl 23.25 < $NHCl_2$ 139.1

300 nm: NH_2Cl 1.08 < $NHCl_2$ 135.72

2.4 Application of Breakpoint Chlorination in Water Reuse Systems

2.4.1 Simulation of Chloramine Species Concentrations in RO Permeate Using a Mathematical Model

Over the past decade, researchers have studied UV/HOCl and UV/chloramine as replacements for UV/H_2O_2. When chloramines are present in the RO permeate of a water-reuse system, added HOCl undergoes breakpoint chlorination with them. Zhang et al. (2019b) found that RO permeate from a California water-reuse facility contained approximately 4 mg/L as Cl_2 chloramines, including 2.5 mg/L as Cl_2 (35.21 uM) NH_2Cl and 1.5 mg/L as Cl_2 (10.56 uM) $NHCl_2$. A typical full-scale UV/HOCl dose is 5-8 mg/L as Cl_2 (70.4-112.7 uM) HOCl.

Kintecus simulations based on Jafvert and Valentine (1992) were conducted at pH 5.8 with background concentrations of 35.21 uM NH_2Cl and 10.56 uM $NHCl_2$, adding 84.5 uM (~6 mg/L) HOCl. Under dark conditions, HOCl and NH_2Cl rapidly decreased over approximately 30 seconds to 1 minute, while a large amount of $NHCl_2$ formed (Figure 2-5a). After about 3 minutes, the concentrations of the species approached equilibrium, showing that $NHCl_2$ is the major product after breakpoint chlorination in RO permeate.

When comparing systems with 84.5 uM (~6 mg/L) versus 112.7 uM (~8 mg/L) HOCl, the concentration changes during the first 30 seconds to 1 minute were very similar. At later times, increasing HOCl did not produce more $NHCl_2$; instead, more HOCl remained in the system and $NHCl_2$ decreased slightly after 4-5 minutes. NCl_3 remained below 1 uM in both systems because it reacts very rapidly with $NHCl_2$ (F12, $k_{F12} = 5.6 \times 10^3\ M^{-1}\ s^{-1}$) and is therefore rapidly consumed after formation.

Figure 2-5. Simulation of species concentration changes over five minutes in pH 5.8 RO permeate containing 35.21 uM NH2Cl and 10.56 uM NHCl2, with (a) 84.5 uM HOCl and (b) 112.7 uM HOCl.

The Kintecus simulations show that the main species involved are HOCl, OCl-, NH2Cl, NHCl2, and NCl3. Because chloramine concentrations change most rapidly during the first five minutes of breakpoint chlorination, the radical-generation rates of these oxidants are important to UV AOP efficiency. Full-wavelength molar-absorptivity analysis shows that each species absorbs differently at different wavelengths. Multiplying the average concentrations at 30 seconds by their molar absorptivities from 230-400 nm gives the absorbance in Figure 2-6.

At 254 nm, the absorbances of HOCl, NH2Cl, and NHCl2 are generally low, while OCl- and NCl3 are near zero. Around 290-300 nm, NHCl2 has much higher absorbance, about 0.011 cm^-1. Higher absorbance indicates greater ability to absorb light at a particular wavelength; therefore NHCl2 is the primary absorbing species in this range.

Figure 2-6. Kintecus simulation of absorbance at 30 seconds of breakpoint chlorination using average concentrations of HOCl, OCl-, NH2Cl, NHCl2, and NCl3 multiplied by their molar absorptivities from 230-400 nm. (a) 84.5 uM HOCl; (b) 112.7 uM HOCl. Conditions: pH 5.8; [NH2Cl] = 35.21 uM; [NHCl2] = 10.56 uM.

2.4.2 Pollutant Degradation by UV254 nm/Breakpoint Chlorination in a Laboratory-Scale Reactor

Aghdam et al. (2021) used the reactor shown in Figure 2-9 to compare UV/HOCl, UV/NH2Cl, breakpoint chlorination alone, and UV/breakpoint chlorination for nitrobenzene and benzoic acid removal. Overall removal decreased in the order UV/breakpoint chlorination > UV/free chlorine > UV/NH2Cl > breakpoint chlorination (Figure 2-7).

Figure 2-7. Degradation of (a) benzoic acid and (b) nitrobenzene by UV254 nm/free chlorine, UV254 nm/NH2Cl, breakpoint chlorination, and UV254 nm/breakpoint chlorination.

Conditions: pH 7; [benzoic acid]0 = 5 uM; [nitrobenzene]0 = 5 uM. Source: Aghdam et al. (2021).

Aghdam et al. also found that, under dark breakpoint chlorination, pollutant degradation changed little after about five minutes (Figure 2-8a), indicating that the concentrations of the participating species had begun to stabilize. UV/breakpoint chlorination increased benzoic-acid removal from 20% to 70% compared with dark breakpoint chlorination, but required 35-40 minutes (Figure 2-8b). This was mainly attributed to the conventional semi-enclosed laboratory reactor, in which the distance between the light source and water surface reduces the intensity reaching the water and requires a longer irradiation time.

Figure 2-8. Effect of initial chlorine dosage on benzoic-acid degradation during (a) dark breakpoint chlorination and (b) UV254 nm/breakpoint chlorination. Conditions: pH 7; [benzoic acid]0 = 5 uM; [N-NH3]0 = 1 mg/L; [HOCl]0 = 7.6, 9.6, 12.6, and 17.6 mg/L as Cl2. Source: Aghdam et al. (2021).

Figure 2-9. Schematic of a conventional semi-enclosed laboratory-scale reactor.

Kwon et al. (2020) investigated the effect of chloramine concentration on 1,4-dioxane removal using a pilot-scale system between laboratory and full scale. Figure 2-10 shows the pilot-scale system, based on a Trojan UVPhox(TM) 08AL20 reactor with eight 250 W low-pressure, high-output mercury-vapor UV lamps installed in a 20 cm diameter stainless-steel chamber. The flow rate was 3.4-6.8 m3/h, giving a UV reactor hydraulic retention time of about 44 seconds. After a process change, the dosing point was allowed to pass through approximately three reactor hydraulic retention times, about 1-3 minutes, before sampling to ensure stable conditions (Kwon et al., 2020).

Figure 2-10. Pilot-scale system schematic. The system includes a UV reactor, flow meter, injection ports, static mixer, online UV transmittance meter, pH meter, and DO meter. The influent to the UV reactor came from the advanced water purification facility effluent of the Orange County Water District in Southern California. Source: Kwon et al. (2020).

The Aghdam and Kwon studies indicate that a pilot-scale AOP can provide 700-1000 mJ/cm2 of irradiation in about one minute, whereas laboratory-scale systems using lower-intensity illumination may require about 30 minutes to achieve the same dose. However, because reaction rates and exposure times differ between pilot and laboratory systems, the oxidant forms remaining after irradiation and the DBP composition generated when HOCl or chloramines react with natural organic matter and inorganic substances may differ.

2.5 Relationship Between Radicals Generated during Breakpoint Chlorination and Pollutant Removal

Based on the structures and properties of six model micropollutants - benzoic acid (BA), nitrobenzene (NB), caffeine, carbamazepine (CBZ), 1,4-dioxane, and N,N-diethyl-m-toluamide (DEET) - and their reaction-rate constants with reactive species that may be generated during photolysis of free chlorine and chloramines, including •OH, •Cl, •ClO, and •Cl2- (Patton et al., 2018; Patton et al., 2022; Zhang et al., 2019a), the six pollutants compete for different radicals. Therefore, selecting pollutants with different reactivities toward •OH, •Cl, •ClO, and •Cl2- provides a useful set of indicators for the study.

For example, nitrobenzene is highly reactive toward •OH (k•OH = 3.9x10^9 M^-1 s^-1) (Aghdam et al., 2021), but there is insufficient literature on its reaction with chlorine radicals to predict its actual degradation in water containing other pollutants. 1,4-dioxane is highly reactive toward •OH (k•OH = 3.1x10^9 M^-1 s^-1) (Patton et al., 2017) but is less reactive toward •Cl (k•Cl = 4.4x10^6 M^-1 s^-1). DEET has high reaction-rate constants toward both •OH (4.6x10^9 M^-1 s^-1) and •Cl (3.8x10^9 M^-1 s^-1) (Sun et al., 2016), so when 1,4-dioxane and DEET are present together, DEET preferentially reacts with •Cl and competes with 1,4-dioxane for •OH.

Caffeine and carbamazepine also react with •ClO and •Cl2-, in addition to •OH and •Cl. Their reported constants include k•OH = 6.4x10^9 M^-1 s^-1 for caffeine and 8.8x10^9 M^-1 s^-1 for carbamazepine; k•Cl = 1.5x10^10 M^-1 s^-1 for caffeine and 5.6x10^10 M^-1 s^-1 for carbamazepine (Guo et al., 2017; Lei et al., 2019; Wang et al., 2016).

The reaction-rate constants indicate that •OH preferentially reacts with caffeine, carbamazepine, and DEET; •Cl preferentially reacts with benzoic acid, caffeine, carbamazepine, and DEET; and •ClO and •Cl2- preferentially react with caffeine and carbamazepine. Literature also indicates that nitrobenzene and benzoic acid are not readily degraded by UV but are highly reactive with radicals, so different UV energy densities do not strongly affect their radical reaction rates (Aghdam et al., 2021).

Water-reuse systems contain many unknown contaminants. Because contaminants compete for different radicals, selecting appropriate target pollutants while considering pollutant-radical rate constants is important for making laboratory results more representative of actual waters. Previous studies have generally analyzed only two or three pollutants and have less often considered systematic competition among multiple pollutants. The six pollutants selected here have different structures and radical reactivities and therefore better represent the range of behavior expected in real water.

Table 2-4. Reaction rate constants of pollutants with four radicals.

Target compound	k•OH (M^-1 s^-1)	k•Cl (M^-1 s^-1)	k•ClO (M^-1 s^-1)	k•Cl2- (M^-1 s^-1)	Reference
Nitrobenzene	3.9x10^9	--	--	--	Aghdam et al., 2021
Benzoic acid	5.9x10^9	1.8x10^10	<3.0x10^6	2.0x10^6	Aghdam et al., 2021
Caffeine	6.4x10^9	1.46-3.87x10^10	1.03x10^8	9.28x10^8	Guo et al.; Lei et al.; Sun et al.
Carbamazepine	8.8x10^9	5.6(+/-1.6)x10^10	9.2x10^7	0.43(+/-0.03)x10^8	Guo et al.; Lei et al.; Wang et al.
1,4-dioxane	3.1x10^9	4.4x10^6	--	3.3x10^6	Patton et al., 2017
N,N-diethyl-m-toluamide	6.7x10^9	3.8x10^9	--	--	Sun et al., 2016

2.6 Formation and Degradation of NDMA under Breakpoint Chlorination

According to Szczuka and Mitch et al. (2020), the concentration of nitrosamines increases significantly after RO permeate in a water-reuse system undergoes UV/HOCl AOP. One proposed NDMA formation pathway under breakpoint chlorination is hydrolysis of $NHCl_2$ to nitroxyl/hyponitrous intermediates, followed by reaction of HNO with HOCl to produce nitrosyl chloride (ClNO), which then reacts with dimethylamine (DMA) to form NDMA (Equations 2.12 and 2.13) (Szczuka et al., 2020).

$NHCl_2 + H_2O \rightarrow HNO / ClNO$ (2.12)

$ClNO + DMA \rightarrow NDMA + HCl$ (2.13)

In 2021, Pham and Fairey et al. re-examined the NDMA pathway proposed by Szczuka and Mitch. They proposed that $NHCl_2$ hydrolyzes to HNO, which then reacts with oxygen to form peroxynitrous acid (HOONO) or peroxynitrite ($OONO^-$) (Equation 2.14). These species then react with DMA to produce NDMA (Equation 2.15). Their work identified another possible NDMA pathway and emphasized the importance of HOONO and $OONO^-$ during UV/HOCl treatment (Pham et al., 2021).

$NHCl_2 + H_2O \rightarrow HNO / NO\bullet \rightarrow HOONO / OONO^-$ (2.14)

$HOONO / OONO^- + DMA \rightarrow NDMA$ (2.15)

Because these two groups of researchers proposed somewhat different theories for NDMA formation, Chuang et al. (2023) further investigated whether the HOCl-HNO pathway proposed by Szczuka et al. was related to reactions between NCl_3 and $NHCl_2$. Their results indicated that NCl_3 plays an important role in increasing NDMA formation during breakpoint chlorination because $NHCl_2$ and NCl_3 can react with each other to generate NDMA, while NCl_3 forms through the reaction of HOCl and $NHCl_2$.

$NHCl_2 + NCl_3 \rightarrow NDMA + \text{other byproducts}$ (2.16)

$NHCl_2 + HOCl \rightarrow NCl_3 + H_2O$ (2.17)

The study proposed two ways to reduce NDMA. First, when Cl2/N molar < 2, the accumulated ClNO concentration after one hour decreases as Cl2/N molar increases (Figure 2-12). This indicates that excess HOCl and OCl- convert NHCl2 to NCl3, lower the NHCl2 concentration, reduce the mutual reaction of NCl3 and NHCl2, and thereby reduce NDMA formation. Second, NHCl2 and NCl3 are readily photolyzed in a UV reactor, which can help control nitrosamine formation during breakpoint chlorination.

Previous studies indicate that RO membranes remove only about 30% of NDMA (Szczuka et al., 2020). NDMA, however, has a molar absorptivity at 254 nm of 1650 M^-1 cm^-1. Although UV/breakpoint chlorination promotes NDMA formation, it can also rapidly photolyze NDMA. Therefore, using an appropriate energy density to control NDMA generated during the process is important, particularly in water-reuse systems intended for potable use (McCurry et al., 2017). This study therefore examines whether breakpoint chlorination with a UVB light source can achieve effects comparable to 254 nm.

Figure 2-12. NDMA concentration after adding 0-225 uM HOCl to water containing 50 uM NH4+ and 7.5 uM Cl-DMA at pH 7 and reacting for one hour. The one-hour cumulative ClNO concentration was also simulated under the same conditions without Cl-DMA using the optimized UF model. Source: Chuang et al. (2023).

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CHAPTER 3. EXPERIMENTAL EQUIPMENT AND METHODS

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3.1 Research Framework

The experimental framework of this study, shown in Figure 3-1, is divided into five stages.

Stage 1: Experimental design and photon-flux measurement

To make the research more representative of practical plant conditions, a commercially available annular stainless-steel reactor was selected. The study used an 8 W UVC254 nm light source and 4 W, 6 W, and 8 W UVB306 nm light sources. Because lamp energy differs with lamp wattage, photon-flux experiments were first conducted to determine the photon energy emitted per unit time (s), expressed as joules (J), at 254 and 306 nm for different lamp powers. This allowed the experimental results to be compared more appropriately.

Stage 2: UV/breakpoint chlorination experiments in pH 5.8 and pH 7 phosphate buffers

Breakpoint chlorination pollutant-degradation experiments were conducted in phosphate-buffered solutions at pH 5.8 and pH 7. UVB irradiation was used to excite chloramine species generated during breakpoint chlorination and produce radicals. The study examined whether UVB can effectively generate radicals for pollutant removal and compared pollutant degradation rates and changes in total chlorine concentration during the reaction.

Stage 3: UV/breakpoint chlorination experiments in synthetic RO permeate

The research target was RO permeate from a wastewater-reuse treatment plant after membrane treatment. Because actual RO permeate is difficult to obtain, synthetic RO permeate was prepared in the laboratory. The study investigated pollutant degradation rates and changes in total chlorine during UV/breakpoint chlorination under different Cl2/N molar ratios, background chloramine concentrations, lamp powers, and reactor reflective-surface materials.

Stage 4: Comparison of NDMA photolysis under UVC254 nm and UVB306 nm

This stage compared the rate of NDMA photolysis and the rate of NDMA formation after reactions between dichloramine and trichloramine under UVC254 nm and UVB306 nm with breakpoint chlorination. The main question was whether NDMA can also be effectively photolyzed under a UVB light source.

Figure 3-1. Experimental framework flowchart.

Stage 1:

- Design of the stainless-steel annular reactor.

- Photon-flux measurement: 8 W UVC254 nm and 4 W/6 W/8 W UVB306 nm.

Experimental conditions: flow rate = 2400 mL/min; total volume = 850 mL; Cl2/N molar ratios = 2, 2.5, 3; sampling times = 0, 1, 2, 3, 4, 5, 6, 8, 10 min.

Stage 2:

- Pollutant degradation and total chlorine analysis under different breakpoint chlorination conditions in 10 mM phosphate buffer at pH 5.8 and pH 7.

Experimental conditions: flow rate = 2400 mL/min; total volume = 850 mL; Cl2/N molar ratios = 1.63, 2.17, 2.72, 3.26; initial [NH2Cl] = 35.21 uM (2.5 mg/L as Cl2); initial [NHCl2] = 10.56 uM (1.5 mg/L as Cl2); sampling times = 0, 1, 2, 3, 4, 5, 6, 8, 10, 15 min.

Stage 3:

- Pollutant degradation and total chlorine analysis under different breakpoint chlorination conditions in synthetic RO permeate.

Stage 4:

- Comparison of UV/breakpoint chlorination, the rate of NDMA photolysis, and the rate of NDMA formation from dichloramine and trichloramine.

Experimental conditions: flow rate = 2400 mL/min; total volume = 850 mL; initial [Cl-DMA] = 1 uM; Cl2/N molar = 2.72; sampling times = 0, 1, 2, 5, 7, 12, 17, 22, 28 min.

3.2 Experimental Apparatus, Chemicals, and Analytical Equipment

3.2.1 Annular Stainless-Steel UV Reactor

The reactor configuration shown in Figure 3-2 was based on the research of Bolton James and Linden Karl (2003), Feng and Smith (2007), Yin et al. (2018), and Kwon et al. (2020). The reactor is a stainless-steel vessel with a reflective interior. The reflective stainless-steel surface increases the optical path length and therefore improves photon utilization. The cylindrical vessel contains a hollow quartz tube with the UV light source inside it. The light sources used were 8 W UVC254 nm and 4 W, 6 W, and 8 W UVB306 nm. The maximum capacity of the stainless-steel vessel is 700 mL.

During experiments, the reactor was installed vertically to ensure uniform mixing. A peristaltic pump transferred the water sample from a beaker into the reactor through the inlet. Once the reactor contained sufficient water, water flowed through the outlet tube back to the beaker, creating a rapid, continuous recirculating reaction system.

For sampling, approximately 150 mL of sample was reserved outside the reactor, representing about 21.5% of the reactor volume. A magnetic stirrer mixed the water outside the reactor. This configuration allowed pollutant concentration changes over time to be studied. The error introduced by reserving sample outside the reactor could be accounted for through calculation.

Figure 3-2. Schematic of the annular stainless-steel UV reactor.

3.2.2 Experimental Chemicals

The chemicals were divided into five groups: photon-flux measurement, total chlorine measurement, pollutant degradation, laboratory synthesis of RO permeate, and NDMA photolysis.

Table 3-1. Chemicals for photon-flux experiments.

Chemical | Purity | Manufacturer

Potassium oxalate | 98.8% | Alfa Aesar

Ammonium ferric sulfate, 12-hydrate | 98.5% | J.T. Baker

Sulfuric acid | 97% | Showa Chemical

1,10-phenanthroline | >99% | Acros Organics

Sodium acetate | >99% | Acros Organics

Table 3-2. Chemicals for UV/breakpoint chlorination total-chlorine experiments.

Sodium hypochlorite (min. 6%) | Duksan

Ammonium chloride | ACS | Macron

Monopotassium phosphate | >99% | Acros

DPD oxalate / N,N-diethyl-p-phenylenediamine | Analytical Grade | Hack

Potassium iodide | >99.5% | Honeywell Fluka

Sodium hydroxide | >=98% | Honeywell

Table 3-3. Chemicals for pollutant degradation experiments.

Nitrobenzene | >99% | Acros

Benzoic acid | 98.0% | Acros

Caffeine | 99.0% | Alfa Aesar

Carbamazepine | >97.0% | TCI Japan

1,4-dioxane | 99.8% | Acros

N,N-diethyl-m-toluamide | 98% | Acros

Methyl tert-butyl ether | HPLC grade, >99.0% | Alfa Aesar

1,4-dioxane-d8 | >99.0% | Aldrich

1,2-dibromopropane | 97% | Sigma Aldrich

Methanol | HPLC grade | J.T. Baker

Ascorbic acid | Reagent grade | Sigma

Table 3-4. Chemicals for synthetic RO permeate.

Methanol | HPLC grade | J.T. Baker

Acetone | HPLC grade | J.T. Baker

Monopotassium phosphate | >99% | Acros

Sodium bicarbonate | D100% | You-Ho (Taiwan)

Hydrochloric acid | 35% | Showa Chemical

Nitric acid | 69.0-70.0% | J.T. Baker

Sodium hypochlorite (min. 6%) | Duksan

Ammonium chloride | ACS | Macron

Table 3-5. Chemicals for Cl-DMA and NDMA analysis.

Sodium hypochlorite (min. 6%) | Duksan

Dimethylamine | 40 wt.% | Acros

N-nitrosodimethylamine in methanol | 100% | AccuStandard

N-nitrosodimethylamine-d6 in dichloromethane | 100% | AccuStandard

Dichloromethane | >=99.9% | Honeywell Fluka

3.2.3 Analytical Equipment

Table 3-6 lists the four analytical instruments used in this study. An Agilent Cary 60 UV-Vis spectrophotometer was used for quantification of monochloramine and dichloramine, photon-flux measurement, and total-chlorine analysis. An Agilent 5977A GC-MS and Agilent Infinity II HPLC were used for target pollutant and NDMA concentration measurements. Detailed conditions are shown in Tables 3-7 to 3-11.

Table 3-6. Analytical instruments.

UV-Vis spectrophotometer | Cary 60 UV-Vis | Agilent

Gas chromatography-mass spectrometry | 7890B GC System / 5977A MSD | Agilent

High-performance liquid chromatography | 1260 Infinity II | Agilent

High-pressure ion chromatography | DIONEX INTEGRION | Thermo Fisher

Table 3-7. HPLC parameters for benzoic acid and nitrobenzene.

Column: Zorbax ODS (Part No. 883952-702)

Flow rate: 1 mL/min

Mobile phase: 10 mM phosphate buffer (pH 3) : methanol (v/v)

0-6 min: 30% : 70%

Table 3-8. HPLC parameters for caffeine and carbamazepine.

Column: Zorbax ODS (Part No. 883952-702)

Flow rate: 1 mL/min

Mobile phase: 10 mM phosphate buffer (pH 7) : methanol (v/v)

0-3 min: 60% : 40%

3-13.5 min: 45% : 55%

13.5-13.6 min: 60% : 40%

Table 3-9. GC-MS parameters for 1,4-dioxane and DEET.

Column: RESTEK RTX-200

Initial temperature / hold time: 40 C, 4 min

Temperature ramp: 60 C/min

Final temperature / hold time: 280 C, 4 min

Table 3-10. HPLC parameters for Cl-DMA.

Column: Zorbax ODS (Part No. 883952-702)

Flow rate: 1 mL/min

Mobile phase: 10 mM phosphate buffer (pH 7) : methanol (v/v)

0-5 min: 30% : 70%

Table 3-11. GC-MS parameters for NDMA.

Column: RESTEK RTX-200

Initial temperature / hold time: 40 C, 5 min

Temperature ramp: 10 C/min to 165 C; 60 C/min to 275 C

Final temperature / hold time: 275 C, 5 min

3.3 Experimental Methods

3.3.1 Photon Flux Measurement Method for the Annular Reactor

According to Goldstein and Rabani (2008), the potassium iodide-potassium iodate method has greater than 99.9% light absorption below 285 nm. According to Lee et al. (2007), the ferrioxalate method is suitable in the 300-390 nm range. Therefore, the ferrioxalate chemical actinometry method was selected for measuring photon flux in the UVB range.

During the experiment, 0.02 M ammonium ferric sulfate dodecahydrate was dissolved in 400 mL of 0.1 N H_2SO_4, and 0.06 M potassium oxalate was dissolved in another 400 mL of 0.1 N H_2SO_4. The two solutions were combined in an aluminum-foil-covered beaker, giving a total volume of 800 mL. In the absence of irradiation, the mixed solution was pumped through the annular stainless-steel reactor at 2400 mL/min. The lamp was then switched on while the pump and timer were started simultaneously. Under 306 nm UVB irradiation, samples were collected every 30 seconds, 1.5 mL each, and 1.5 mL of 0.1% (w/v) 1,10-phenanthroline solution was added. Samples were allowed to react in the dark for 30 minutes, after which absorbance of $Fe(1,10\text{-}phen)3^{2+}$ was measured at 510 nm using a UV-Vis spectrophotometer. The emitted photon energy per unit time (J/s) was then calculated.

3.3.2 Micropollutant Degradation Method

To investigate the effects of water pH and Cl2/N molar on pollutant degradation, 10 mM phosphate-buffer solutions at pH 5.8 and pH 7 were prepared. Because ascorbic acid undergoes a redox reaction with HOCl and deactivates chlorine, ascorbic acid could be used before sample analysis to terminate breakpoint chlorination and ensure accurate pollutant measurements.

Experimental procedure:

1. Warm up the UV lamp for 30 minutes.

2. To 850 mL of 10 mM phosphate buffer at pH 5.8 or pH 7, add 0.5 uM each of nitrobenzene, benzoic acid, CBZ, and caffeine, and 0.2 uM each of 1,4-dioxane and DEET. Mix thoroughly, then add 50 uM NH4Cl.

3. With the lamp off, pump the mixed solution through the annular stainless-steel reactor at 2400 mL/min.

4. Add 100, 125, or 150 uM HOCl to the beaker outside the reactor, corresponding to Cl2/N molar = 2, 2.5, or 3. Immediately after adding HOCl, switch on the lamp and start the timer.

5. Collect samples at specified times and store them in 20 mL vials containing ascorbic acid at 1.1 times the total chlorine concentration.

Total chlorine was analyzed during the experiment using the N,N-diethyl-p-phenylenediamine (DPD) colorimetric method, based on APHA Standard Method 4500-Cl G.

Analysis procedure:

1. Add 150 uL of 385 mM pH 6.5 phosphate buffer to 3 mL of sample.

2. Add 150 uL of 3.81 mM DPD indicator.

3. Add approximately 1 g of potassium iodide powder.

4. Mix thoroughly and allow to react for 2 minutes.

5. Measure absorbance at 515 nm using a UV-Vis spectrophotometer.

After the experiment, pollutant concentrations were analyzed. Different instruments required the following procedures:

1. Transfer 1 mL of the 10 mL sample to a small vial and analyze nitrobenzene, benzoic acid, CBZ, and caffeine using HPLC-UV.

2. Add 300 uL of 0.514 uM 1,4-dioxane-d8 to the remaining 9 mL as the internal standard for 1,4-dioxane.

3. Add 2 mL methyl tert-butyl ether and 12 g Na2SO4, shake, and allow to stand for 2 minutes for liquid-liquid extraction (LLE).

4. Transfer the upper organic layer to a small vial using a glass dropper and analyze 1,4-dioxane and DEET by GC-MS.

3.3.3 Preparation of Synthetic RO Permeate

To investigate RO permeate from actual water-reuse facilities, which is difficult to obtain, experiments were conducted using laboratory-synthesized RO permeate. The synthetic water was then used for UV/breakpoint chlorination AOP experiments at different chlorine-to-nitrogen molar ratios.

The composition of the synthetic RO permeate was mainly based on Mangalgiri et al. (2019) and Zhang et al. (2019a). Literature indicates that RO permeate in potable-reuse systems commonly contains carbonate (~0.5 mM), DOC (~0.2 mg/L), chloramines (28.2-42.3 uM NH2Cl and 9.2-14.1 uM NHCl2), chloride, and inorganic nitrogen species that scavenge radicals, such as NO3-. The study therefore added these components to 2 mM phosphate-buffered deionized water at pH 5.6 to prepare synthetic RO permeate. Literature reports a k•OH of 2.1-2.5x10^4 m (mg-C/L)^-1 s^-1 for DOC in RO permeate, so a mixture of 77% acetone and 23% methanol on a molar-carbon basis was used to represent the DOC contribution. The resulting synthetic RO permeate had k•OH of approximately 2.1x10^4 (mg-C/L)^-1 s^-1.

Individual solutions were prepared according to the composition in Table 3-12. To 5 L of DI water, the prepared sodium bicarbonate solution, hydrochloric acid, nitric acid, DOC solution, phosphate buffer, 35.21 uM NH2Cl, and 10.56 uM NHCl2 were added in sequence. The pH of the 5 L mixture was finally adjusted to 5.6 using 5% phosphoric acid.

Table 3-12. Synthetic RO permeate parameters.

Parameter | Value

Phosphate buffer (mM) | 2

Sodium bicarbonate (mM) | 0.5

Hydrochloric acid (mM) | 0.17

Nitric acid (mg/L-N) | 1

DOC | 23% methanol + 77% acetone

NHCl2 (uM) | 1.3-2.0

NH2Cl (uM) | 2.0-3.0

pH | 5.6

3.3.4 UV AOP Experiments Using Synthetic RO Permeate

To investigate the effects of UVB306 nm under different chlorine-to-nitrogen molar ratios, background chloramine concentrations, and experimental conditions (90-second dark reaction before irradiation, reduced lamp wattage, and changed reactor reflective-surface material) on pollutant degradation, synthetic RO permeate at pH 5.6 was prepared.

Experimental procedure:

1. Warm up the UV lamp for 30 minutes.

2. Add 0.5 uM nitrobenzene, benzoic acid, CBZ, and caffeine, and 0.2 uM 1,4-dioxane and DEET, to 850 mL of synthetic RO permeate and mix thoroughly.

3. With the lamp off, pump the solution through the annular stainless-steel reactor at 2400 mL/min.

4. Add 53.52, 90.13, 126.75, or 163.37 uM HOCl to the beaker outside the reactor, approximately corresponding to Cl2/N molar = 1.63, 2.17, 2.72, and 3.26. Immediately after adding HOCl, switch on the lamp and start the timer.

5. Collect samples at specified times and store them in 20 mL vials containing ascorbic acid at 1.1 times the total chlorine concentration.

6. Analyze total chlorine during the experiment and pollutant concentrations after the experiment.

3.3.5 NDMA Degradation Method

To compare NDMA formation and degradation rates under UVC254 nm and UVB306 nm, synthetic RO permeate at pH 5.6 and Cl2/N molar = 2.72 was used. According to Szczuka et al. (2020), quenching agents such as ascorbic acid, sodium thiosulfate, and sodium sulfite react with Cl-DMA and therefore are unsuitable for Cl-DMA analysis. After repeatedly confirming that chloramines do not react with Cl-DMA over a short period, glycine was used to quench the HOCl reaction.

Experimental procedure:

1. Warm up the UV lamp for 30 minutes.

2. Add 1 uM Cl-DMA to 850 mL of synthetic RO permeate and mix thoroughly.

3. With the lamp off, pump the solution through the annular stainless-steel reactor at 2400 mL/min.

4. Add 126.75 uM HOCl to the external beaker (Cl2/N molar = 2.72), immediately switch on the lamp, and start the timer.

5. Collect samples at specified times and store them in vials containing ascorbic acid or glycine at 1.1 times the Cl-DMA concentration.

6. Analyze Cl-DMA by HPLC-UV during the experiment.

After the experiment, NDMA concentration was analyzed after concentration and purification pretreatment.

Analysis procedure:

1. Add 30 uL of 10 mg/L N-nitrosodimethylamine-d6 to 15 mL of sample as the NDMA internal standard.

2. Add 4 mL dichloromethane (DCM) and shake rapidly for 2 minutes, then allow the phases to separate for 2 minutes for liquid-liquid extraction.

3. Transfer the lower DCM layer with a glass dropper to a screw-cap tube containing approximately 1 g Na2SO4 to remove residual water.

4. Transfer the upper organic solvent from the tube to another clean tube and purge with nitrogen, concentrating it to approximately 500 mL.

5. Transfer the concentrated solution to a small vial and analyze NDMA by GC-MS.

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CHAPTER 4. RESULTS AND DISCUSSION

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4.1 Photon Flux Results for the Annular Reactor

The lamps used in the study were: (1) 8 W, 254 nm; (2) 4 W, 306 nm; (3) 6 W, 306 nm; and (4) 8 W, 306 nm. Photon-flux experiments were first performed to determine the photon energy emitted per unit time in the stainless-steel annular reactor.

Photon flux was determined by measuring the molar absorptivity of Fe(1,10-phen)3^2+ at 510 nm and calculating its complex concentration using epsilon510 nm = 11,100 M^-1 cm^-1. Plotting complex concentration against time gives a linear relationship. The slope of concentration versus time was used to calculate photons emitted per second. Multiplying the photon emission rate by the energy contained in one mole of Einstein photons at the

specified wavelength gives the photon flux, expressed as photon energy emitted per unit time (J/s) (Equations 4.1-4.2).

Photon energy emitted (J/s) = (Slope x V / QY) x U (4.1)

where V is solution volume (L); QY is quantum yield. According to Goldstein and Rabani (2008), QY = 1.40 for UVC254 nm and 1.25 for UVB306 nm.

U(lambda) = (hc/lambda) x 6.02 x 10^23 x 10^3

= (6.626x10^-34 x 2.998x10^8 / lambda) x 6.02x10^23 x 10^3 (4.2)

where U is the energy contained in one mole of Einstein photons at the specified wavelength (mJ/mole Einsteins); h is Planck's constant (6.626x10^-34 m2 kg/s); and c is the speed of light (2.998x10^8 m/s).

The ferrioxalate method was used to measure photon flux from UVC254 nm and UVB306 nm lamps of different wattages. The calculated photon energies emitted per unit time are summarized in Table 4-1 and compared with manufacturer information (Figures 4-1 and 4-2).

Table 4-1. Calculated photon flux for UVC254 nm and UVB306 nm.

	8W UVC254	4W UVB306	6W UVB306	8W UVB306
No. of photons emitted (E/s)	3.15x10^-6	7.68x10^-7	2.48x10^-6	4.16x10^-6
Photon energy emitted (J/s)	1.71	0.45	0.97	1.63
Manufacturer value (J/s)	1.80	0.50	1.00	1.60

Figure 4-1. Philips TUV 8W G8T5 lamp information.

Source: PChome product listing.

Figure 4-2. SANKYO DENKI UVB G4T5E lamp information.

Source: SANKYO DENKI UV-B lamp technical document.

4.2 Degradation of Micropollutants by Breakpoint Chlorination without Irradiation in Phosphate Buffer

4.2.1 Control Experiments under HOCl, NH4+, and Irradiation Conditions

The study first examined pollutant degradation in phosphate buffer after adding HOCl, NH4+, or irradiation. As shown in Figure 4-3, pollutant concentrations did not change significantly under these conditions.

Figure 4-3. Experimental control groups. In 10 mM, pH 7 phosphate buffer, the six pollutants were tested under: (a) no irradiation, no HOCl, 50 uM NH4Cl; (b) no irradiation, 75 uM HOCl, no NH4Cl; and (c) 8 W UVB306 nm, no HOCl, 50 uM NH4Cl. Error bars represent the range of duplicate measurements.

4.2.2 Breakpoint Chlorination Degradation of Pollutants in 10 mM, pH 5.8 Phosphate Buffer

Literature indicates that small amounts of radicals are still produced during breakpoint chlorination in the dark. Therefore, before studying UV/breakpoint chlorination, dark breakpoint chlorination experiments were conducted with nitrobenzene and 1,4-dioxane to ensure that the pollutants would not be extensively removed during breakpoint chlorination itself.

The reaction of 0.5 uM nitrobenzene and 0.2 uM 1,4-dioxane in 10 mM, pH 5.8 phosphate buffer was observed over 15 minutes. Figure 4-4 shows that at Cl2/N molar = 2, pollutant concentrations changed more noticeably than at the other ratios, but after 3-5 minutes the

concentrations changed little. These results confirm that radicals produced by dark breakpoint chlorination degraded only about 5% of the pollutants.

Figure 4-4. Dark breakpoint chlorination degradation results for (a) nitrobenzene and (b) 1,4-dioxane in 10 mM, pH 5.8 phosphate buffer. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 100, 125, and 150 uM; [NH4Cl] = 50 uM. Error bars represent the range of duplicate measurements.

4.3 Comparison of UV/Breakpoint Chlorination and UV/H2O2 for Degradation of Model Pollutants in Phosphate Buffer

UV/H2O2 is a mature AOP commonly used in water treatment plants for controlling organic pollutants in drinking-water sources and RO units in water reuse (Kwon et al., 2020). However, H2O2 has low photon-utilization efficiency and may remain in the treated effluent, increasing disinfection costs.

The initial hypothesis of this study was to use a laboratory-scale stainless-steel reactor to conduct UVC254 nm/breakpoint chlorination and UVB306 nm/breakpoint chlorination pollutant-degradation experiments, using 8 W UVC254 nm with 3 mg/L H2O2 as the comparison baseline to determine whether UV/breakpoint chlorination could increase AOP efficiency.

In kinetic studies, ln([C]/[C]0) = -kobs x t is commonly used to describe first-order reactant concentration changes. When ln([C]/[C]0) varies linearly with time, the slope is -kobs, where kobs is a pseudo-first-order degradation rate constant with units of s^-1 or min^-1. It can be used to compare pollutant degradation. Because lamps of different wattages produce different photon fluxes during the same irradiation time, direct comparison by time is not appropriate. Therefore, experimental results were plotted as ln([C]/[C]0) versus photon energy emitted per unit time (J/s).

4.3.1 Comparison of UVB306 nm/Breakpoint Chlorination under Weakly Acidic and Neutral Conditions

Previous studies indicate that when pH changes from weakly acidic to neutral, intermediates formed by oxidant or free-chlorine decomposition can increase radical scavenging and lower pollutant-removal efficiency in UV/breakpoint chlorination AOPs (Chuang et al., 2017; Zhang et al., 2019b). Therefore, phosphate buffers at pH 5.8 and pH 7 were used to investigate the effect of pH on pollutant degradation during breakpoint chlorination under irradiation.

Experiments were first conducted using 8 W UVB306 nm with Cl2/N molar = 2 to degrade nitrobenzene and 1,4-dioxane at different pH values, and the results were compared with 8 W UVC254 nm and 3 mg/L H2O2. For nitrobenzene, kobs at pH 5.8 and pH 7 under 8 W UVB306 nm/breakpoint chlorination were 0.0019 and 0.0028 J^-1, respectively. For 1,4-dioxane, kobs values were 0.0013 and 0.0025 J^-1, respectively.

Under weakly acidic conditions (pH 5.8), pollutant degradation removal was more than 32% greater than under neutral conditions (pH 7) for both nitrobenzene and 1,4-dioxane, consistent with the literature. In addition, under weakly acidic conditions, 8 W UVB306 nm/breakpoint chlorination produced pollutant-removal performance about 1.2 times that of 8 W UVC254 nm/H2O2. This indicates feasibility of UVB306 nm/breakpoint chlorination in a weakly acidic environment.

Figure 4-5. Comparison of degradation of (a) nitrobenzene and (b) 1,4-dioxane at pH 5.8 and pH 7, with Cl2/N molar = 2, and with 3 mg/L H2O2, under 8 W UVB306 nm and 8 W UVC254 nm irradiation. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 100 uM; [NH4Cl] = 50 uM. Error bars represent the range of duplicate measurements.

4.3.2 Comparison of UVB306 nm/Breakpoint Chlorination at Different Chlorine Dosages under Weakly Acidic Conditions

After examining pH effects, experiments were conducted in weakly acidic pH 5.8 phosphate buffer to investigate how different Cl2/N molar ratios affect pollutant degradation. An 8 W UVB306 nm source was used with Cl2/N molar ratios of 2, 2.5, and 3

to degrade benzoic acid, nitrobenzene, caffeine, carbamazepine (CBZ), 1,4-dioxane, and N,N-diethyl-m-toluamide (DEET), and kobs values were compared.

Figure 4-6 shows the results. When photon energy emitted per unit time was plotted against ln([C]/[C]0), benzoic acid, nitrobenzene, and 1,4-dioxane all showed clearly poorer degradation at Cl2/N molar = 2 than at 2.5 or 3. All six pollutants had larger kobs values at Cl2/N molar = 3.

At Cl2/N molar = 3, the kobs values for benzoic acid, nitrobenzene, CBZ, caffeine, 1,4-dioxane, and DEET were 9.2x10^-3, 6.6x10^-3, 9.4x10^-3, 1.4x10^-2, 4.4x10^-3, and 7.7x10^-3 J^-1, respectively. Nitrobenzene, 1,4-dioxane, and DEET had relatively smaller kobs values and therefore poorer removal. A possible reason is that these three pollutants have lower reactivity with the •ClO and •Cl2- radicals generated during breakpoint chlorination. This interpretation is supported by the reported rate constants for reactions of the pollutants with •OH, •Cl, •ClO, and •Cl2-.

In weakly acidic buffer, comparison of 8 W UVB306 nm/breakpoint chlorination at different chlorine-to-nitrogen ratios with 8 W UVC254 nm/H2O2 showed that although H2O2 can generate large amounts of •OH and all six pollutants are highly reactive toward •OH, increasing Cl2/N molar from 2 to 2.5 or 3 substantially improved pollutant degradation and could exceed removal by 8 W UVC254 nm/H2O2.

Figure 4-6. Degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET in 10 mM, pH 5.8 phosphate buffer after 8 W UVB306 nm irradiation at different Cl2/N molar ratios. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 100, 125, and 150 uM; [NH4Cl] = 50 uM. Error bars represent the range of duplicate measurements.

4.4 Comparison of Different AOPs for Removal of Model Micropollutants from Synthetic RO Permeate

Experiments in phosphate buffer showed that weakly acidic conditions and increased chlorine dosage helped degrade model micropollutants. The next experiments therefore investigated the effects of different chlorine dosages and UV wavelengths on six model micropollutants in laboratory-synthesized RO permeate.

4.4.1 Breakpoint Curve of Synthetic RO Permeate

According to Mangalgiri et al. (2019) and Zhang et al. (2019a), RO permeate contains carbonate (~0.5 mM), DOC (~0.2 mg/L), chloramines (28.2-42.3 uM NH_2Cl and 9.2-14.1 uM $NHCl_2$), chloride, and inorganic nitrogen species such as NO_3^- that scavenge radicals.

In this study, the ammonia nitrogen and chlorine sources in synthetic RO permeate were NH_2Cl and $NHCl_2$. NH_2Cl was prepared by slowly adding 5 mL of 40 mM HOCl to 5 mL of 48 mM NH_4Cl and mixing thoroughly (Schreiber and Mitch, 2005). $NHCl_2$ was prepared by diluting the NH_2Cl solution to 3.5 mM and adjusting the pH to 3.7-3.8 with 5% phosphoric acid, maintaining that pH for about 10 minutes, and then placing the solution in an ice bath at 4 C for one hour (Chuang et al., 2023).

When 35.21 uM NH_2Cl and 10.56 uM $NHCl_2$ were selected as the background chloramine concentrations, the corresponding total nitrogen and total chlorine concentrations contributed by these species were calculated to be 67.6 uM and 56.3 uM. Because total nitrogen divided by total chlorine gives a Cl_2/N molar ratio of 0.83, additional NH_4Cl would be required to make Cl_2/N molar approach 0 or 0.5 (Figure 4-7).

Because the composition of synthetic RO permeate is complex, the breakpoint position may differ from the theoretical Cl_2/N molar = 1.5 reported by Metcalf et al. (2007). To determine the breakpoint of the synthetic RO permeate, different amounts of HOCl were added and total chlorine was measured at different Cl_2/N molar ratios. Figure 4-7 shows that the breakpoint occurred at Cl_2/N molar = 2-2.25. Subsequent experiments therefore used Cl_2/N molar ratios of 1.63, 2.17, 2.72, and 3.26 to investigate pollutant degradation by UVC254 nm and UVB306 nm AOPs.

Figure 4-7. Breakpoint chlorination curve for synthetic RO permeate measured after 30 minutes of reaction at different Cl2/N molar ratios. Background chloramine concentrations: 35.21 uM NH2Cl and 10.56 uM NHCl2; these contribute 67.6 uM total nitrogen and 56.3 uM total chlorine; Cl2/N molar = 0.83. Additional NH4Cl is needed to reach Cl2/N molar = 0 or 0.5. Error bars represent the range of duplicate measurements.

4.4.2 Total Chlorine Changes and Pollutant Degradation under 8 W UVC254 nm/Breakpoint Chlorination

To investigate whether the short-term changes in chloramine concentrations during UV/breakpoint chlorination could be effectively utilized with an appropriate light source to improve pollutant degradation, 8 W UVC254 nm was used with different chlorine dosages to treat synthetic RO permeate containing the six model micropollutants.

First, total chlorine concentrations in the system were measured under different chlorine dosages during treatment with 8 W UVC254 nm. As shown in Figure 4-8, there was no obvious difference in total chlorine degradation among the different chlorine dosages. However, the pollutant degradation results in Figure 4-9 showed that the energy-normalized degradation rates (energy-normalized kobs) were faster at Cl2/N molar = 3.26. This was mainly because the chloramine species in the synthetic RO water and the added HOCl could generate more radicals under UV irradiation. The greater radical production reduced the time and energy required for pollutant degradation and produced better removal.

Figure 4-8. Total chlorine concentration changes in synthetic RO permeate after irradiation with 8 W UVC254 nm at Cl2/N molar ratios of 1.63, 2.17, 2.72, and 3.26. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 53.52, 90.13, 126.75, and 163.37 uM. Error bars represent the range of duplicate measurements.

Figure 4-9. Degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET in synthetic RO permeate after irradiation with 8 W UVC254 nm at different Cl2/N molar ratios. Conditions: 0.5 uM each of benzoic

acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 53.52, 90.13, 126.75, and 163.37 uM. Error bars represent the range of duplicate measurements.

4.4.3 Total Chlorine Changes and Pollutant Degradation under 8 W UVB306 nm/Breakpoint Chlorination

Next, to investigate the feasibility of using UVB306 nm with different chlorine-to-nitrogen molar ratios for pollutant degradation, an 8 W UVB306 nm source, the same wattage as the UVC254 nm source described above, was used for subsequent AOP experiments.

The changes in total chlorine concentration were examined at different chlorine dosages (Figure 4-10). The largest change occurred at Cl2/N molar = 2.72, followed by Cl2/N molar = 3.26. The pollutant degradation results showed the same trend (Figure 4-11).

For example, for 1,4-dioxane, the kobs values at Cl2/N molar ratios of 1.63, 2.17, 2.72, and 3.26 were 1.0x10^-3, 2.6x10^-3, 4.3x10^-3, and 3.8x10^-2 J^-1, respectively. Based on kobs, increasing the chlorine dosage to Cl2/N molar = 2.72 improved 1,4-dioxane degradation. At this chlorine-to-nitrogen ratio, 1,4-dioxane removal was about 35% higher than that obtained with 8 W UVC254 nm/H2O2 (kobs = 2.8x10^-2 J^-1). During breakpoint chlorination, HOCl and chloramine species generate radicals such as •OH, •Cl, •ClO, and •Cl2- under UV irradiation. Changes in species concentrations over time also affect radical production. In particular, adding HOCl beyond the breakpoint substantially increased the efficiency of the 8 W UVB306 nm/breakpoint chlorination AOP.

Figure 4-10. Total chlorine concentration changes in synthetic RO permeate after irradiation with 8 W UVB306 nm at Cl2/N molar ratios of 1.63, 2.17, 2.72, and 3.26. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 53.52, 90.13, 126.75, and 163.37 uM. Error bars represent the range of duplicate measurements.

Figure 4-11. Degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET in synthetic RO permeate after irradiation with 8 W UVB306 nm at different Cl2/N molar ratios. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 53.52, 90.13, 126.75, and 163.37 uM. Error bars represent the range of duplicate measurements.

4.4.4 Comparison of Pollutant Degradation by UVC254 nm/H2O2 and UV/Breakpoint Chlorination

After conducting pollutant degradation experiments with 8 W UVC254 nm and 8 W UVB306 nm at different chlorine dosages, the results at Cl2/N molar = 2.72, above the breakpoint, were compared further with UVC254 nm/H2O2. The oxidant concentration changes in Figure 4-12 and pollutant degradation results in Figure 4-13 show that both UVC254 nm/breakpoint chlorination and UVB306 nm/breakpoint chlorination improved pollutant degradation by 1.6-2.7 times compared with conventional UVC254 nm/H2O2. This result highlights the advantages and potential future applicability of UV/breakpoint chlorination.

However, a direct comparison of 8 W UVC254 nm/breakpoint chlorination and 8 W UVB306 nm/breakpoint chlorination showed that the UVC254 nm/breakpoint system had 20%-50% higher changes in total chlorine concentration and higher degradation rates for benzoic acid, nitrobenzene, 1,4-dioxane, and DEET. A possible reason is that, under 254 nm and 306 nm irradiation, changes in species concentrations over time cause the species to absorb the specific light source differently, generating different types and concentrations of radicals and therefore different pollutant degradation rates. The subsequent experiments therefore sought to improve photon utilization by dichloramine through optimization of the experimental conditions, thereby generating higher radical concentrations and increasing their reaction with pollutants.

Figure 4-12. Changes in oxidant concentrations in synthetic RO permeate during treatment with 8 W UVC254 nm/H2O2, 8 W UVC254 nm/breakpoint chlorination, and 8 W UVB306 nm/breakpoint chlorination at Cl2/N molar = 2.72. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [H2O2] = 3 mg/L; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

Figure 4-13. Comparison of degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET in synthetic RO permeate after treatment with 8 W UVC254 nm/H2O2 and UV/breakpoint chlorination at Cl2/N molar = 2.72. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [H2O2] = 3 mg/L; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

4.5 Effect of Different Chloramine Concentrations on AOP Treatment Performance

The synthetic RO permeate used in the previous experiments contained 35.21 uM NH2Cl and 10.56 uM NHCl2. However, Zhang et al. (2019b) reported that the background chloramine concentrations in the RO permeate of a California water reuse facility were 28.2-42.3 uM NH2Cl and 9.2-14.1 uM NHCl2. This study therefore further investigated the effect of different background chloramine concentrations in RO permeate on AOP treatment performance.

The chloramine concentrations in the synthetic RO permeate were increased to 49.30 uM NH2Cl and 14.58 uM NHCl2, 1.4 times the initial NH2Cl and NHCl2 concentrations. Figures 4-14 and 4-15 show that the kobs values for the total chlorine concentration change and for the six model micropollutants were nearly identical with and without the increased chloramine concentration. Thus, at the same Cl2/N molar ratio, increasing the background chloramine concentration of synthetic RO permeate did not significantly affect pollutant degradation.

Figure 4-14. Total chlorine concentration changes after irradiation with 8 W UVB306 nm at Cl2/N molar = 2.72 in synthetic RO permeate containing either [NH2Cl] = 35.21 uM and [NHCl2] = 10.56 uM or [NH2Cl] = 49.30 uM and [NHCl2] = 14.58 uM. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

Figure 4-15. Degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET at Cl2/N molar = 2.72 after 8 W UVB306 nm

irradiation in synthetic RO permeate containing either [NH2Cl] = 35.21 uM and [NHCl2] = 10.56 uM or [NH2Cl] = 49.30 uM and [NHCl2] = 14.58 uM. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

4.6 Optimization of UVB306 nm/Breakpoint Chlorination AOP Operating Conditions

The initial comparison of different wavelengths and chlorine dosages showed that both wavelength and whether the light energy could be effectively utilized by chloramine species affected pollutant degradation. The following experiments therefore modified the conditions for UVB306 nm/breakpoint chlorination at the optimal chlorine dosage identified in the pollutant degradation experiments, Cl2/N molar = 2.72: (1) a 90-second dark reaction after chlorination before irradiation; (2) reducing the energy supplied per unit time by the UVB306 nm lamp; and (3) changing the reactor reflective surface material. These experiments were used to investigate optimal operating conditions for UVB306 nm/breakpoint chlorination AOP.

4.6.1 Effect of a 90-Second Dark Reaction after Chlorination on Pollutant Degradation

Using the mathematical model developed by Jafvert and Valentine (1992) in Kintecus, the study previously simulated synthetic RO permeate at pH 5.8 containing 35.21 uM NH2Cl and 10.56 uM NHCl2 with 126.75 uM HOCl. During the dark reaction after chlorination, HOCl and NH2Cl rapidly decreased and a large amount of NHCl2 formed. Based on this model, the average concentrations of HOCl, OCl-, NH2Cl, NHCl2, and NCl3 at 10, 30, 60, and 90 seconds of breakpoint chlorination were multiplied by their molar absorptivity coefficients at different wavelengths to obtain the results in Figure 4-16.

The simulation in Figure 4-16 shows that from 10 to 90 seconds, at 290-300 nm the absorbance of HOCl gradually decreased while the absorbance of NHCl2 increased markedly. NHCl2 had its highest absorbance at 90 seconds. Therefore, the experiment selected a 90-second dark reaction after chlorine addition followed by irradiation as a potential UVB306 nm/breakpoint chlorination AOP operating condition.

Figure 4-16. Kintecus simulation of the average concentrations of HOCl, OCl-, NH2Cl, NHCl2, and NCl3 and the resulting absorbance obtained by multiplying their concentrations by their molar absorptivity coefficients from 230 to 400 nm at (a) 10 s, (b) 30 s, (c) 60 s, and (d) 90 s of breakpoint chlorination. Conditions: pH 5.6; [NH2Cl] = 35.21 uM; [NHCl2] = 10.56 uM; [HOCl] = 126.75 uM.

The central question was whether the concentrations of HOCl and NHCl2 in the breakpoint chlorination system, which change with reaction time, affected the efficiency of UV AOP pollutant degradation. Based on the simulation, synthetic RO permeate was prepared with 126.75 uM HOCl (Cl2/N molar = 2.72), allowed to react in the dark for 90 seconds, and then irradiated with 8 W UVB306 nm.

However, Figure 4-17 shows that performing a 90-second breakpoint chlorination reaction before irradiation did not support the original hypothesis. Instead, degradation of the six model micropollutants was 17%-34% lower than with direct UVB306 nm/breakpoint chlorination. A possible reason is that HOCl can directly generate •OH and •Cl after irradiation. Although the 90-second breakpoint reaction creates a higher NHCl2 concentration, NHCl2 requires a longer irradiation period to generate radicals. Therefore, the concentrations of both HOCl and NHCl2 that can generate radicals need to be considered simultaneously. Performing the 90-second breakpoint chlorination reaction before irradiation was therefore not an effective method for improving UVB306 nm/breakpoint chlorination AOP treatment performance.

Figure 4-17. Degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET after a 90-second dark reaction in synthetic RO permeate at Cl2/N molar = 2.72 followed by irradiation with 8 W UVB306 nm. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [NH2Cl] = 35.21 uM; [NHCl2] = 10.56 uM; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

4.6.2 Effect of Reducing Lamp Energy Supplied per Unit Time on Pollutant Degradation

Considering the changes in the concentrations of the different species during breakpoint chlorination, the species in the system compete for absorption of light energy. Once the UV source is turned on, oxidant concentrations change significantly compared with the dark condition. This experiment therefore further investigated whether reducing the light energy supplied per unit time could create a higher concentration of dichloramine in the system.

Experiments were conducted with 4 W, 6 W, and 8 W UVB306 nm sources in synthetic RO permeate at Cl2/N molar = 2.72. Figure 4-18 shows that reducing the lamp power from 8 W to 4 W increased the standardized kobs for total chlorine concentration change from 2.7x10^-3 J^-1 to 3.9x10^-3 J^-1, indicating that lower lamp power increased the change in total chlorine concentration in the system. The pollutant degradation results in Figure 4-19 similarly showed that reducing the lamp power from 8 W to 4 W increased the energy-normalized degradation rates for nitrobenzene and 1,4-dioxane by 18% and 13%, respectively.

This result was consistent with the original hypothesis and indicates that reducing the light energy supplied per unit time can create more NHCl2 in the system. When dichloramine absorbs 306 nm light, it can greatly increase its absorbance and generate more radicals (•Cl), thereby increasing the ability of the UV AOP to degrade pollutants in water. If excessively high light intensity is supplied immediately after HOCl addition, HOCl may absorb the light and directly generate •OH, which may reduce the concentration of dichloramine produced and may also reduce the ability of dichloramine to absorb light energy.

Figure 4-18. Total chlorine concentration changes after irradiation of synthetic RO permeate at Cl2/N molar = 2.72 with 4 W, 6 W, and 8 W UVB306 nm sources. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

Figure 4-19. Degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET in synthetic RO permeate at Cl2/N molar = 2.72 after irradiation with 4 W, 6 W, and 8 W UVB306 nm sources. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

4.6.3 Effect of Changing Reactor Reflective Surface Material on Pollutant Degradation

The reactor used in this study had a stainless steel reflective surface. According to the literature (Kim et al., 2004), stainless steel has a light reflectance of 20%-30%. Haefner et al. (2017) reported that Teflon (polytetrafluoroethylene, PTFE) has excellent light reflectance of 90%-100%. Because reflective surface material may affect the amount of light reaching the water per unit time and therefore affect dichloramine photon utilization, the study tested UV/breakpoint chlorination using reactors with three reflective surface materials: glass, stainless steel, and Teflon. Figure 4-20 shows the reactor dimensions and the different reflective materials. For the Teflon reactor, a 0.5 mm thick Teflon sheet was attached to the inside of a glass reactor (Figures 4-20b and 4-21).

Teflon is composed of repeating tetrafluoroethylene monomers. The highly electronegative fluorine atoms produce strong molecular bonding, reducing surface defects and absorption and increasing light reflectance. Using Teflon in lamps or reflector panels can improve photon utilization efficiency and reduce energy consumption. Glass, with a light reflectance of 4%-8%, is transparent, strong, chemically resistant, and easy to clean and is widely used in buildings, household products, electronic products, and transportation. However, because its surface is relatively rough and its refractive index is high, its ability to reflect light is relatively poor.

Figure 4-20. (a) Reactor dimensions, (b) Teflon reactor, and (c) glass reactor schematic.

Figure 4-21. Information on the Polytetrafluoroethylene (PTFE) film. Source: https://www.sigmaaldrich.com/TW/en/product/aldrich/gf73473701

In photochemical reactions, the effective path length of light can be determined by measuring the rate of H_2O_2 photodegradation in order to examine the distribution of light energy in a reactor. Effective path length differs from the physical straight-line distance because it incorporates the combined effects of multiple reflections, scattering, and absorption of light in the medium. In systems with highly scattering or highly reflective media, repeated scattering or reflection increases the effective path length.

Following the experimental method used by Garoma and Gurol (2005) to measure effective path length, 0.05 mM H2O2 and 1 mM tert-butyl hydroperoxide were added to 700 mL of 1 mM, pH 7 phosphate buffer, with tert-butyl hydroperoxide used as a scavenger for hydroxyl radicals (•OH). The prepared solution was placed in stainless steel and Teflon reactors and irradiated with 8 W UVC254 nm or 8 W UVB306 nm. Samples were collected over time and H2O2 concentrations were measured. Plotting ln([H2O2]/[H2O2]0) against time gave the first-order reaction rate constant kobs.

Figure 4-22 shows the path-length experiment results for stainless steel and Teflon reactors under 8 W UVC254 nm and 8 W UVB306 nm irradiation. Under 8 W UVC254 nm, the Teflon reactor had an H2O2 degradation kobs of 7.01x10^-2 min^-1 compared with 1.48x10^-2 min^-1 for the stainless steel reactor, an increase of about 78.9%. Under 8 W UVB306 nm, the Teflon reactor kobs was 2.9x10^-3 min^-1 compared with 6.0x10^-4 min^-1 for stainless steel, an increase of about 79.3%. These results demonstrate that Teflon can increase the effective optical path length and light reflectance and provide a basis for the subsequent experiments examining reflective surface materials.

Figure 4-22. Relationship between H2O2 concentration and irradiation time in 700 mL, 1 mM, pH 7 phosphate buffer containing 1 mM tert-butyl hydroperoxide and 0.05 mM H2O2 under (a) 8 W UVC254 nm and (b) 8 W UVB306 nm irradiation. Error bars represent the range of duplicate measurements.

Based on the previous experiments on irradiation after breakpoint chlorination and reduction of the energy supplied per unit time, the best UVB306 nm/breakpoint chlorination AOP operating condition was 4 W UVB306 nm with Cl2/N molar = 2.72, with irradiation started immediately after chlorine addition. The present experiment therefore investigated whether changing the reactor reflective surface material and increasing reflectance could improve UV AOP pollutant degradation.

Figure 4-23 shows total chlorine concentration changes for glass, stainless steel, and Teflon reflective surfaces. The standardized kobs values were 3.6x10^-3 J^-1, 4.5x10^-3 J^-1, and 8.3x10^-3 J^-1 for glass, stainless steel, and Teflon, respectively. The results show that using a Teflon reactor markedly increased the change in total chlorine concentration.

Further analysis of pollutant degradation (Figure 4-24) showed that the Teflon reactor required only about 200-250 J of light energy to degrade approximately 98% of the pollutants. Compared with the stainless steel reactor, degradation rates for benzoic acid, nitrobenzene, caffeine, carbamazepine, 1,4-dioxane, and DEET increased by 50.6%, 56.5%, 34.0%, 34.2%, 53.9%, and 53.3%, respectively. Thus, using the highly reflective Teflon surface can effectively reduce photon-energy loss in the system while increasing dichloramine photon utilization per unit time.

Figure 4-23. Total chlorine concentration changes after irradiation of synthetic RO permeate at Cl2/N molar = 2.72 with a 4 W UVB306 nm source using glass, stainless steel, and Teflon reactor reflective surfaces. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

Figure 4-24. Degradation of (a) benzoic acid, (b) nitrobenzene, (c) caffeine, (d) carbamazepine, (e) 1,4-dioxane, and (f) DEET in synthetic RO permeate at Cl2/N molar = 2.72 after irradiation with 4 W UVB306 nm using different reactor reflective surfaces. Conditions: 0.5 uM each of benzoic acid, nitrobenzene, CBZ, and caffeine; 0.2 uM each of 1,4-dioxane and DEET; [HOCl] = 126.75 uM. Error bars represent the range of duplicate measurements.

4.6.4 Comparison of Optimized UVB306 nm/Breakpoint Chlorination AOP and UV/H2O2 AOP

The main objective was to identify chlorine dosage and operating conditions that create a higher concentration of dichloramine in synthetic RO permeate, so that UVB irradiation can interact with dichloramine to generate more radicals that react with and remove pollutants. However, when dichloramine favors radical-generation pathways, it may reduce the likelihood of forming trichloramine and the disinfection byproduct NDMA.

The experiments identified the following optimal conditions for UVB306 nm/breakpoint chlorination AOP: a 4 W UVB306 nm source in a Teflon reactor, Cl2/N molar = 2.72, with irradiation started immediately after chlorine addition. Table 4-2 summarizes the energy-

normalized degradation rates for the three AOPs tested. Comparing 4 W UVB306 nm at Cl2/N molar = 2.72 in a PTFE reactor with 8 W UVC254 nm/H2O2 in a stainless steel reactor, the kobs values for benzoic acid, nitrobenzene, caffeine, carbamazepine, 1,4-dioxane, and DEET increased by 66.4%, 73.0%, 82.7%, 68.0%, 72.5%, and 67.0%, respectively. Comparing the optimized 4 W UVB306 nm + Cl2/N molar = 2.72 PTFE condition with 8 W UVC254 nm + Cl2/N molar = 2.72 in stainless steel, the corresponding kobs values increased by 25.1%, 27.0%, 56.5%, 65.6%, 29.4%, and 47.6%, respectively.

Table 4-2. Energy-normalized pollutant degradation rates for different AOPs.

Pollutant | 8 W UVC254 nm/H2O2 with stainless steel | 8 W UVC254 nm + Cl2/N molar = 2.72 with stainless steel | 4 W UVB306 nm + Cl2/N molar = 2.72 with PTFE

Benzoic acid | 0.0079 | 0.0176 | 0.0235

Nitrobenzene | 0.0031 | 0.0084 | 0.0115

Caffeine | 0.0060 | 0.0151 | 0.0347

Carbamazepine | 0.0117 | 0.0126 | 0.0366

1,4-dioxane | 0.0028 | 0.0072 | 0.0102

DEET | 0.0075 | 0.0119 | 0.0227

Units: J^-1.

The comparison indicates that optimized UVB306 nm/breakpoint chlorination AOP improved advanced oxidation removal efficiency. It therefore has potential for future application of wastewater reuse for potable water. To increase the feasibility of full-scale implementation, future work should further investigate whether optimized UVB306 nm/breakpoint chlorination generates more disinfection byproducts or causes greater toxicity.

4.7 Effect of UV/Breakpoint Chlorination on NDMA in Synthetic RO Permeate

According to Szczuka et al. (2020) and Kwon et al. (2020), NDMA can be generated during advanced oxidation treatment of RO permeate in water-reuse systems. However, NDMA can be rapidly photolyzed at 254 nm (Soltermann et al., 2013). Because this study focuses on 306 nm irradiation, additional experiments were conducted to investigate NDMA formation and degradation under UVC254 nm and UVB306 nm irradiation.

Synthetic RO permeate was spiked with 1 uM Cl-DMA as a reaction precursor. Irradiation experiments were conducted at Cl2/N molar = 2.72 in an 8 W UVC254 nm stainless steel reactor, a 4 W UVB306 nm stainless steel reactor, and a 4 W UVB306 nm Teflon reactor. Figure 4-25a shows that regardless of the light source, when approximately 140 J of energy had been supplied, Cl-DMA had mostly reacted with other species in the water to form NDMA or had been photolyzed.

The NDMA concentration results in Figure 4-26 show several observations. First, the NDMA concentration generated by the 8 W UVC254 nm source was about half that generated by the 4 W UVB306 nm source, and approximately 300 J of energy was sufficient to completely photolyze NDMA. Second, the 4 W UVB306 nm stainless steel reactor took the longest to reach the maximum NDMA formation concentration. NDMA photolysis in this reactor required approximately 300 J more energy than in the 8 W UVC254 nm stainless steel reactor and approximately 600 J more than in the 4 W UVB306 nm Teflon reactor, corresponding to 1-2 times more energy. Third, although the NDMA photolysis rate in the 4 W UVB306 nm Teflon reactor was lower than in the 8 W UVC254 nm stainless steel reactor, the excellent reflectance of Teflon made both NDMA formation and photolysis approximately twice as fast as in the 4 W UVB306 nm stainless steel reactor. Fourth, with the 8 W UVC254 nm/H2O2 comparison condition, NDMA was photolyzed immediately as it was formed, so NDMA was not detected in the analysis.

According to Kwon et al. (2020), California water-reuse regulations require NDMA concentrations in the effluent after advanced oxidation treatment to be below 10 ng/L and require at least 0.5 log removal of 1,4-dioxane, corresponding to approximately 68% removal. Considering both NDMA and 1,4-dioxane concentrations, the 4 W UVB306 nm Teflon reactor required 300 J more light energy than the 8 W UVC254 nm stainless steel reactor. This result can provide a reference for future studies analyzing other disinfection byproducts and toxicity in treated water.

Figure 4-25. Changes in (a) Cl-DMA and (b) NDMA concentrations with irradiation energy in synthetic RO permeate containing 1 uM Cl-DMA as a reaction precursor at Cl2/N molar = 2.72. Conditions: 8 W UVC254 nm and 4 W UVB306 nm sources; stainless steel and Teflon reactor reflective surfaces; [HOCl] = 126.75 uM; [H2O2] = 3 mg/L. Error bars represent the range of duplicate measurements.

CHAPTER 5. CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

This study investigated the effects of UVB306 nm/breakpoint chlorination AOP on pollutant degradation and tested optimal operating conditions. The following conclusions were obtained:

1. The steady-state concentration of radicals has a major effect on pollutant removal. Compared with neutral conditions, weakly acidic conditions (pH 5.8) help increase radical stability and thereby improve pollutant removal by UV AOP.

2. After adding different HOCl concentrations to synthetic RO permeate and measuring total chlorine, the breakpoint occurred at Cl2/N molar = 2-2.25. Experiments with different chlorine dosages under 8 W UVB306 nm/breakpoint chlorination showed that chlorine addition above the breakpoint improved pollutant removal. In particular, at Cl2/N molar = 2.72, similar pollutant degradation rates to other chlorine-to-nitrogen ratios could be achieved with less energy.

3. Increasing the background chloramine concentration in synthetic RO permeate did not significantly affect pollutant degradation.

4. Experiments investigating optimal operating conditions for UVB306 nm/breakpoint chlorination AOP showed that reducing the light energy supplied per unit time can create

more dichloramine in the system. When dichloramine absorbs 306 nm light, more radicals (•Cl) can be generated. Using highly reflective Teflon as the reactor reflective surface also reduces photon-energy loss and increases dichloramine photon utilization per unit time. Therefore, breakpoint chlorination with 4 W UVB306 nm in a Teflon reactor can increase the ability of UV AOP to degrade pollutants in water.

5. Comparison of UV/H2O2 AOP, 8 W UVC254 nm/breakpoint chlorination AOP, and optimized UVB306 nm/breakpoint chlorination AOP, together with the pollutant degradation and NDMA photolysis results, indicated that breakpoint chlorination with 4 W UVB306 nm in a Teflon reactor has potential for future wastewater reuse applications for potable water. However, application requires further consideration of whether UVB306 nm/breakpoint chlorination AOP generates more disinfection byproducts or causes greater toxicity.

5.2 Recommendations

1. The experiments in this study involved changes in many nitrogen-containing species, including ammonia nitrogen, chloramine, nitrate nitrogen, and nitrite nitrogen. Total nitrogen (TN) is an important water-quality parameter that can be used to evaluate the nutrient status and degree of pollution of a water body. Future experiments are therefore recommended to use a TOC instrument to measure changes in TN.

2. Future research can further investigate whether UVB306 nm/breakpoint chlorination AOP promotes the formation of nitrogen-containing disinfection byproducts compared with UVC254 nm/breakpoint chlorination AOP.

3. During wastewater reuse treatment, oxidants inevitably react with organic and inorganic substances in the water and produce disinfection byproducts (DBPs). These DBPs can cause ecological toxicity, persistence, and bioaccumulation and may pose developmental, acute, and carcinogenic risks to human health. Before UVB/breakpoint chlorination AOP is applied at full scale, further experiments should analyze additional DBP species and conduct toxicity tests to determine whether more DBPs are generated or greater toxicity occurs.

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